



System Acceptance Review
BRACU Mongol-Tori
BRAC University
University Rover Challenge - 2022



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1. Introduction: Human nature has always been driven by the desire to know the unknown. Following this, our neighboring red planet has always been an enigmatic destination for us, and in hopes of exploring more about it, BRAC University students embarked on an exhilarating adventure with BRACU Mongol-Tori, a next-generation Mars rover, capable of assisting astronauts as well as doing scientific research activities. The team has already performed numerous significant improvements to the rover's overall structure over the past few years. This year we are Introducing “Mongol-Tori Encephalon” with a redesigned and improved version of the Artificial intelligence emphasized subsystems, performing the tasks autonomously in line with the specifications for URC 2022. This year, we provided the rover a new appearance by implementing substantial improvements to the autonomous, network, control, and mechanical systems, as well as fully redesigning the science setup and electrical systems from scratch. Minor adjustments include modifying the chassis, upgrading the arm and claw, and modernizing the wheel. Following the specifications of URC 2022, this report analyzes the Core Rover Systems, Approach to Competition Tasks, Testing and Operations, Team Budget, Gantt Chart, and Science Task.

2. Core Rover Systems: Using our experience of previous URCs, the virtual competition in 2021, and several trials and errors since the competition, we have brought a few significant changes in the mechanical setup, network system, electronics, and science module.

2.1 Mechanical: Our mechanical setup has significant changes to its design after the experience in the virtual competition. We are using the previous four-wheeler setup with remarkable modifications to the wheel. The claw of the rover also has a major update compared to our previous design.



Figure- 1(a) Full body rover 1(b) Modified extender 1(c) New wheel 1(d) Arm with new claw

2.1.1 Chassis and Suspension: The rover is 0.7m long and 0.55m wide with an H-shaped ladder chassis for holding the circuit box and arm. To reduce deformation while moving through rocky terrains, a triangulation is installed in the chassis. The heavy space frame transmits the heavy load as tension and compression load along the length of each strut. Our suspension system is the modified rocker-bogie suspension system for a four-wheeler setup, where two bodies are connected directly to the chassis, with the U-shaped differential bar at the rear part of the system to give extra support. There are two universal joints in the differential bars for magnifying the rotational angle of bogies, where our feedback actuators and the length bogies can turn from 1.2m to 0.76m. The extender suspension is used to ensure stability during the vertical drops. This year, a custom-made double-sided linear actuator is used in the extender, which reduces the weight of the rover. A planetary gearbox with a custom design has been installed for the custom-made actuator. Also, a bending mechanism has been added for adjusting the height of the rover. **Figure 1(a)** shows our full working rover's design. **Figure 1(b)** shows the use of our modified extender. The extender increases the length of its base and prevents flipping over during vertical drops.

2.1.2 Wheel: The wheels are constructed with a stainless steel ring via a U-shaped rim, have a high breaking point with a 0.3-meter diameter and 0.10-meter width. The design of the shock-absorbing ability makes the wheel stronger, with small holes ruptured to make it lighter. The rubber pads in the

wheels are used to rectify 360-degree rotation and deal with a vertical drop. However, this year, jute fiber ropes are added to our wheels for easy climbing and to increase traction as shown in **figure 1(c)**.

2.1.3 Arm: A strong 360-degree arm is constructed with six degrees of freedom (base, shoulder, elbow, wrist) with two degrees of freedom in the end effector as shown in **figure 1(d)**. Inverse kinematics algorithms have been implemented in our arm using feedback actuators. A chain and sprocket system in the wrist and worm gear at the base of the rover has been implemented. The worm gear locks the arm in place to prevent rotation if the motor fails. The three-finger claw is lightweight, strong, and has been designed to ensure greater precision with a better grip, which can be used for various tasks. The custom drill under the wrist helps break down the soil. For more control and precision, we have used the angle values for inverse kinematics. Additionally, we have used Rviz and Gazebo for the visualization of our arm's position and movement. This year a Scara arm is being developed for our science project.

2.2 Network and Vision: The modified and upgraded communication system of this year includes two dedicated sub-systems, independent of each other. Both the 2.4GHz system and the 5.8GHz system have detached power distribution units from the core rover system. For seamless and continuous data flow in between the rover and base station, a peer-to-peer network paradigm is employed with two high-end 2.4GHz routers. In the base station, a directional sector antenna is paired with a Ubiquiti Rocket M2 Base Station. The height-adjustable antenna stand allows us to stay under the 3-meter height restriction. For rotating the antenna up to 360 degrees, we use a rotator controlled by an elevation azimuth controller. In the rover, a 2.4GHz 10dBi 2x2 high gain MIMO omnidirectional antenna (TL-ANT2410MO) is integrated with another Ubiquiti Rocket M2 router to attain up to 360-degree signal coverage and more than 1-kilometer smooth rover control. A PoE network switch connects the Rover's computing unit to the Rocket M2 router. This includes two IP cameras as well, that are placed in strategic locations throughout the rover, which enables us to provide power and transmit data at the same time. However, with the increasing range, there is a high possibility of facing lag using the 2.4GHz system. The 5.8GHz analog vision system is employed to overcome this issue. Four analog cameras with adjustable 2.5 mm lenses equipped with two pairs of 800mW transmitters and receivers coupled with polarized 3dbi mushroom antennas are set around the rover for a consistent view. Two of these cameras can be accessed simultaneously, while the other two can be accessed through a dual-channel analog switcher on the control system, and thus provide a desirable view with the adjustable pan-tilt mechanism.

2.3 Control System: The control system of the rover will be controlled from a portable base station that will be operated using 1 laptop and 3 displays for camera vision, with a joystick, gamepad, and keyboard.

2.3.1 Electronics: The electronics subsystem this year has been redesigned entirely by dividing the whole system into different parts having different purposes. Starting with the power management board. The board controls and maintains the whole rover's power distribution from the primary source which is the rechargeable Lithium-Ion Battery(BL1845) providing 18 V and 10Ah current supply. The powerline goes through the kill-switch mechanism and passes through MOSFET protection. Also, this board shows the current flow and voltage measurements both onboard and in GUI, in the base station. Then, there is a divider board that has been arranged with the microcontrollers. The arm actuators are being controlled by Dual Monster Moto Shield VNH3ASP30 DC Motor Driver and the wheels are being controlled by Cytron 30A Bi-Directional DC Motor Driver. Furthermore, there will be a manual & debug board for helping to debug the driver board's problems and to allow controlling the rover manually onboard. Moreover, there is a controller board having the developer boards like Arduino, ESP32, Raspberry Pi, and Jetson Nano to connect with the whole system. The circuit (PCB) of this year's electronics system is shown in **figure 2(c)**.

2.3.2 Software: This year we have implemented a Robot Operating System (ROS) to control the rover and use ROS Melodic on Ubuntu 18.04 LTS. As the main control architecture, Raspberry 4 is being used. This

time, the master node is executed on the base station using ROS network configuration. On the base station, a publisher ROS node delivers commands to the rover, while on the rover, a subscriber ROS node receives orders from the base station. For more precision and convenience, the whole rover is now controlled by the Gamepad. MQTT has been utilized as the primary communication channel. A bidirectional MQTT bridge is being utilized to provide capabilities between ROS and MQTT and to improve communication. In addition, inverse kinematics has been introduced, and a package called 'Moveit' has been used to assist in improved arm control and movement. PyQt5 is used to create a new graphical user interface that allows the rover driver to quickly control the rover and view critical information shown in **figure 2(a)**. It may be used to regulate the rover's speed, determine the angle of different joints on the arm, and analyze the circuit connection while it is operating. With the help of the GUI, the voltage can be checked again, and the orientation of the rover, including the 3D presentation, can be inspected. A supplementary control system was also implemented employing Python socket programming. Both ROS system and Socket system separately use the same GUI. Again, different tabs for the missions have been introduced. The feedback system is one of the essential changes that we have continued from last year. To use feedback end effectors and feedback mechanisms, we created a feedback receiving environment in our control GUI. In addition, we used PyQt5 to create the science GUI that can interpret sensor values into custom measurements and display graphical patterns on the screen, making it easy to identify distinct observations. In addition, data from sensors is recorded in a CSV file for further study. The atmospheric data analysis is provided from the sensor to the graph on the science tab shown in **figure 2(d)**

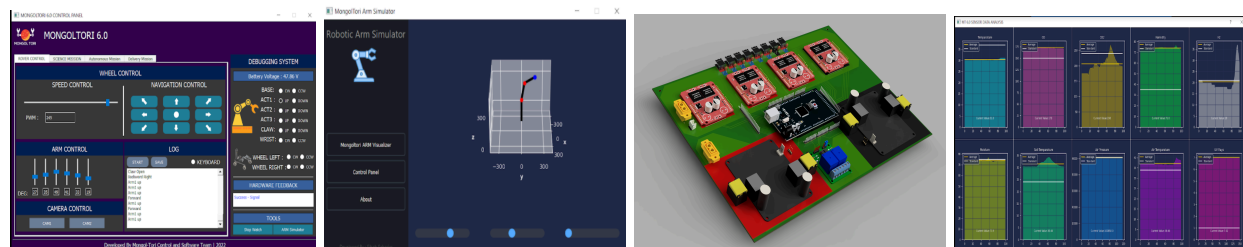


Figure- 2(a)Rover Control GUI 2(b)Arm Simulation 2(c) Circuit System 2(d) Sensor Data Analysis

3. Approach to Competition Mission: To perform the tasks in URC, we have performed trial tasks on locations similar to the competition grounds within the allotted time.

3.1 Extreme Retrieval and Delivery Mission: Our modified rocker-bogie suspension and the strong chassis are fitting for carrying out the extreme retrieval and delivery mission. The rover is designed to be able to move easily up to 1 kilometer and the suspension system can absorb the sudden jolts while traversing through rocky terrain. The new, modified wheel gives a stronger grip over sandy and rough areas due to jute fiber rope. Our expandable rover body allows the rover to retain its stability during the vertical drop. The durable arm is effectively able to lift heavy objects like toolboxes and other equipment. An AI model is trained for mechanical tools classification like a hammer, toolbox, screwdriver, and other equipment using TensorFlow object detection API. The model selects interest areas, then feeds them to a convolutional neural network that generates feature maps. The maps are classified using a support vector machine, where we perform regression between predicted and ground truth bounding boxes. We used OpenCV and a Sharp distance sensor to get the distance of the tools.

3.2 Equipment Servicing Mission: This year, we have used a three-finger claw. The new claw helps us grip with greater control and precision, which will substantially improve our performance during the task. The feedback end effector has been added for accurate movements and the angle is measured using inverse kinematics. In this mission, the artificial Intelligence System assists the astronaut by

detecting elements like Toggle Switch, Joystick, Keyboard, First Aid Box, and instructing the driver for the mission accordingly. Light AI models are used in Jetson Nano for classification and detection.

3.3 Autonomous Traversal Mission: For this task, again NVIDIA Jetson Nano is used which processes the whole Autonomous Navigation System connected with an Arduino Mega through serial communication. Arduino Mega, on the other hand, is also connected to peripheral devices. The Navigation System consists of four individual sensors. The MPU9250 IMU sensor achieves the compass data of the rover and assists the rover to rotate and position itself towards the direction of individual waypoints. For precise localization, Sparkfun GPS-RTK2 Board ZED-F9P differential GNSS module has been used. Due to the pandemic scenario, the RTK GPS is not in our hands yet, so in the meantime, our regular GPS modules have been utilized to implement differential algorithms to develop our custom DGPS system which results in better accuracy. A 2D RPLiDar A1 is used to map the surrounding environment where it localizes obstacles and finds the shortest path for the rover to reach the waypoints. The rover searches the AR tags by rotating up to a 90-degree angle on each right and left side individually. For Distance Estimation of AR Tags, OpenCV is used. After calibrating the camera, we got our desired equation, which resembles a 3rd-degree polynomial equation that is used in the code to find out the desired distance. For Street Signs Identification YOLO v5 object detection algorithm has been used. For vision data, Logitech C505 USB performs significantly. Additionally, the rover is capable of determining when to blink the particular LEDs. While traversing through the waypoints both single ones and between gates, the rover is able to detect AR tags using the 4x4_50 dictionary of the OpenCV ARUCO library. Our AR-tag detection program can easily detect and identify the underlying information very smoothly from different angles and light contrasts.

4. Safety Measures: To ensure the rover's safety measurement 'kill switch' mechanism has been used. This can compel the rover's systems to shut down immediately in the case of an emergency. Also, a MOSFET-based protection circuit has been added to the main circuit to assure the safety of the main circuit board.

5. Testing and Operation: To assure that the rover functions well, Mongol-Tori follows a three-step testing procedure.

a. Sub-System Testing: Initially, for the subsystem testing, the wheels, suspensions, chassis, sample collector, science box, camera positioning were designed in the Solidworks program and evaluated through a simulation from 10/09/2021 to 25/12/2021. After that designs were applied and tested. The mechanical structure of the rover was tested from 2/2/2022 to 19/2/2022 and network and communication setup was tested from 9/2/2022 to 19/2/2022. The GUI which is adaptable to any platform was assessed on 30/10/2021.

b. Complete System Testing: The rover has undergone a lot of modifications since the previous year's URC feedback. This year, several changes have been introduced in each subsystem. Now, this year after assembling the system, we have a complete rover body and mind emphasizing AI. The full rover system testing and development have been completed within 28/2/2022 and a full rover with full setup has been tested on 1/3/2022.

c. Operational Testing: The operational testings were started on 13/2/2022. An autonomous test was carried out on 17/02/2022 at a site with full GPS accessibility. Again, on 19/02/2022 extreme retrieval tasks were conducted. Moreover, on 23/02/2022, a custom setup has been used to accomplish a variety of equipment service tasks, such as turning a switch on and off, turning knobs, collecting cache boxes, undoing latches, and so on. The science module is designed to identify indications of life, execute microbial biomass testing, capillary water flow testing, and amino acid testing, among other things. The rover was intended to gather soil samples utilizing the sample collector on 28/02/2022 to perform the onboard testing and science task.

Science Task: For this mission, we have divided our plan into two parts. One part of our science module is the atmospheric analysis, and another is soil analysis. These Scientific tests are being carried out to determine whether life has ever existed on Mars and to study the planet's environment in preparation for future human exploration. Our rover has formed into an autonomous planet analyzer over the past few years. We have created a sensor box for atmospheric analysis that can function as an onboard laboratory. The onboard sensor box (shown in figure-3(a)) contains Air Temperature and Air Humidity Sensor (DHT22), Sensor-MQ135 (Air Quality Monitoring. Sensitive to benzene, alcohol, smoke, Carbon dioxide), H2 Sensor (MQ8), Light Intensity Measurement Sensor (LDR), UV Light & Radiation Measurement Sensor (ML8511), Oxygen sensor (groove oxygen sensor), Sensor BMP-280 (Barometric pressure sensor 360-degree Panorama View of Arena), pH sensor, microscopic camera, panoramic shot(**figure-3(c)**). Environmental data are collected by these sensors and sent to the base station. These sensor outputs provide valuable climate, soil, and landscape information. A graphical user interface (GUI) for science is designed to assist us in analyzing and delivering our collected data to the control system. For a panoramic shot, we took a unique method where we used a servo-based pan-tilt IP camera and a unique homography matrix-based algorithm to take three images from three different angles and transform them into a panoramic image (**ratio 1:3**).

Moreover, This year we have implemented AI in our science module. The rover's microscopic camera examines the texture of the rocks, and the AI model classifies and categorizes them. For this, we have used the YOLOv5 algorithm, which is based on PyTorch and Convolutional Neural networks. Now, we are working on weather forecasting, and we have used data from the Rover Environmental Monitoring Station (REMS) on the Curiosity Rover to figure out what the weather will be like on Mars—then translated terrestrial columns to date for temperature prediction. Due to the regression problem, we employed LSTM, a Gated Recurrent Neural Network.

Another part of our science module is soil analysis. Our soil analysis section is separated into two sections. We used probes to measure soil NPK (Nitrogen, Phosphorus, and Potassium), soil moisture, and soil temperature. To collect the data from these probes, we used Modbus programming through an RS-485 transceiver. Another segment is our soil testing box. We have made a circular pyramid-shaped box which is shown in **figure 3(b)**.

There are three portions in the box, and each portion can test Biomass, Water capillary, Amino acid, Starch, respectively. A load cell detects weight variation, and a nichrome wire heater generates heat for the Biomass test. The measurement for Biomass = $(\text{weight difference}/\text{previous weight}) \times 100$. We have made a customized test tube for the water capillary test where water will be stored in the lower part. When the soil is dropped into the test tube, the soil absorbs the water, and a moisture sensor detects the water absorbance level in the soil. To identify the presence of amino acid in the sample, we used a ninhydrin reagent and a nichrome wire to generate heat in the test tube. The ninhydrin solution changes color if the soil contains the amino acid. For detecting Starch, we are using iodine solution, which can indicate the existence of Starch in soil. By performing all these analyses our system will be able to find the past footprint of life, indicate the possibility of life in the future, and analyze the environmental change in Mars.

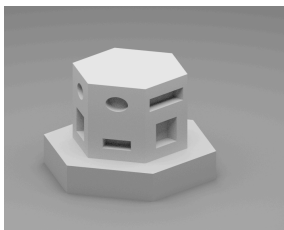
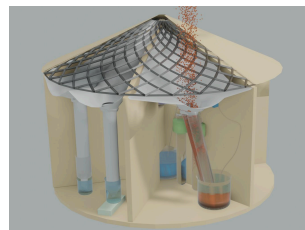


Figure- 3(a) Sensor Box



3(b) Science Box,



3(c) Panoramic Picture